(1992) to the V. Rajagopal Rao panel (2020), all recommended various steps to overhaul the DRDO's structure and steps to greater indigenisation of latest technologies.

As India celebrates the success of ISRO's indigenously developed space technology, DRDO is criticised for being unable to rid India of the unenviable distinction of being the world's largest arms importer. However, though detractors might harp on its tardiness, DRDO can be rightly proud of its missile and radar technologies. The DRDO's foremost successes are the Agni cruise missiles, Tejas and Arjun. Along with Pralay surface-to-surface missile and the Pinaka rocket system, they have been inducted into the armed forces.

While India's defence establishment believes that the DRDO should focus on futuristic air, ground, maritime and space systems, a majority of the DRDO officials say that, for years, its laboratories were tasked to meet the immediate needs of the military. Some military analysts agree and point out that most projects of the defence manufacturing sector are limited to the production of outdated tanks, guns, helicopters, weapons platforms or the setting up of manufacturing units to produce foreign weapons through technology transfers. Now, the terms of reference of the VijayRaghavan-led committee have recast the DRDO's main objective to the development of latest technology. Lt Gen. D.S. Hooda (retd), the former northern army commander, claims that the country lags in cutting-edge technology and the adoption of Artificial Intelligence, robotics, quantum technology and hypersonic weapons. "The focus should be on these high-end technologies on which future wars are going to be fought," he says.

Most defence scientists welcome the government panel, but add that it should not undermine DRDO's contribution. However, A.K. Singh, the former director general of DRDO, admits with candour that the organisation could not adapt to change over the years. "It requires a proper diagnosis and treatment rather than symptomatic treatment," Singh says, while vouching for the capability of the DRDO scientists. The DRDO's approach, he adds, lost vital focus on defence research as its canvas of activities became wider—a case of simultaneous pursuit of too many projects. For example, the DRDO has a Directorate of Civil Works & Estates, which looks after the construction of government buildings. Singh claims such pursuits show how earlier committees set up to address DRDO's problems failed to provide solutions.

MANY COMMITTEES

The most recent committee to look into defence research was the five-member panel led by V. Rajagopal Rao of IIT Delhi set up by the Ministry of Defence (MoD) in 2020 to overhaul the DRDO to meet "defence and battlefield needs". Its recommendations were never made public and only led to the closure of some laboratories. Before that, a panel headed by P. Rama Rao, former secretary of the department of science and technology, submitted a report on the DRDO in

A MIXED TRA

DELAYED PROJECTS

NAVAL LIGHT COMBAT AIRCRAFT (LCA, NAVY)

Scheduled date of completion **March 2010**

Status: First prototype landed on INS *Vikrant* in February 2023

Budget: ₹ **7,000 cr.**

The indigenous combat jet for our aircraft carriers. It is being developed in two variants—single-seater and twin-seat



ADVANCED TOWED ARTILLERY GUN SYSTEM (ATAGS)

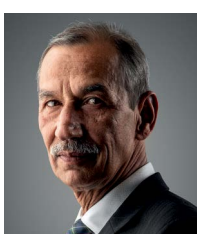
Scheduled for **Mar. 2017**

Status: Army will soon induct over 300 ATAGS

Budget: ₹ **290 cr.**



ATAGS is a large calibre 155/ 52 mm towed gun system that can programme and fire future Long Range Guided Munitions (LRGMs). Developed to replace the army's ageing artillery pieces. After a long gestation process, the defence acquisition council in March 2023 gave its nod for procuring the ATAGS



"THERE IS A NEED TO HARNESS AND SYNERGISE THE EFFORTS OF THE DRDO, DPSUs, THE PRIVATE SECTOR R&D CENTRES AND ACADEMIA, AND THE CHANGE HAS TO BE PUSHED BY THE GOVERNMENT"

LT GEN. D.S. HOODA (RETD)
Former commander, Northern Command